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A SENSITIVITY STUDY ON THE IMPACT OF PRESURISATION OF BUNTER SANDSTONE FORMATION WITHIN THE LICENCE CS028 ON BRINE MIGRATION AND SALINITY CHANGE.

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DECLARATION

I, VALENTINE NAYEBALE confirm that this work submitted for assessment is my own and is expressed in my own words. Any uses made within it of the works of other authors in any form (e.g., ideas, equations, figures, text, tables, programs) are properly acknowledged at the point of their use. A list of the references employed is included.

Signed: Valentine Nayebale

Date: 18th August 2025

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First and foremost, I would like to greatly thank the Almighty God, who has brought me this far, provided me with the knowledge, good health, and the motivation to complete this project and my master's degree in petroleum engineering at Heriot-Watt University. This journey would not have been possible without his endless protection and guidance in my life.I want to express my sincere appreciation to my supervisor, Professor Erick Mackay, for his boundless guidance and support throughout this project. His continuous advice and encouragement truly motivated me to study hard and work smart on this project. I am genuinely grateful for the time and effort he dedicated to mentoring me and consider it a privilege to have had the opportunity to learn under his supervision. Special thanks also go to Dr. Tim Good for his assistance in contributing to the progress of this project. I appreciate all the professors in the Institute of GeoEnergy Department for their efforts in delivering course materials, as I gained valuable knowledge that I believe will help me contribute positively to future projects run by energy companies in the industry. I want to thank my family, especially my parents, who sponsored me throughout my entire master's academic journey. I am very grateful to them for providing emotional and financial support during this time. I appreciate my siblings and friends for their support during the tough times of the examination period and other challenging moments of the course. Finally, I want to thank the coursemates with whom I worked and studied throughout this course. They were a fantastic team to collaborate with, especially on the group field development project and with those I teamed up with for the Petrol Bowl competition in France. I attained valuable teamwork skills from these experiences, which will be a great asset moving forward in my career journey.

EXECUTIVE SUMMARY

This research involves the evaluation of the pressurization of the Bunter Sandstone Formation (BSF) located within the license CS028 area in the southern North Sea. The Bunter Sandstone Formation (BSF) is a monoclinal, gently dipping, Triassic saline aquifer that runs from York to Nottingham and outcrops on the eastern coast of England. This study mainly examines how far brine can migrate up-dip and change the salinity of the fresh waters during large-scale injection of CO_2. The potential risk of brine leakage through the outcrop is assessed, as well as crossflow into the shallow freshwater aquifers located in the overlying LLF. The upward migration of brine could result from legacy wells, which could create a leakage pathway for brine. (Litang Hu,et al, 2012) into the Lincolnshire Limestone Formation (LLF). Any oil well drilled in the 1940s into the underlying reservoir possesses a high risk of brine leakage, as its current integrity and isolation from the Bunter and Lincoln formations are not certain (A.Kopp et al, 2010). A series of sensitivities using a compositional numerical simulation model was analyzed to investigate the influence of boundary conditions and formation geometry on brine migration and resulting salinity changes. Minimal brine displacement is observed as pressure builds up within the Bunter, with displacement only occurring in the upper layers of the formation. This suggests that the less dense CO_2 plume primarily moves in the top layers of the Bunter Sandstone Formation (BSF), displacing brine in those regions. Salinity changes within the Bunter Sandstone Formation (BSF) are small due to the limited brine displacement. High salinity changes are observed in the (LLF) following the flow of brine from the leaky well, as the Bunter Sandstone Formation (BSF) becomes over-pressured as a consequence of low transmissibility of the boundaries. The more closed the boundaries are, the higher the volume of brine flowing into the Lincolnshire Limestone Formation (LLF). The farthest migration front with the (LLF) is observed 1.5 km from the proximity of the leaky well modelled. No brine breakthrough to the Bunter outcrop is noted, as the brine does not migrate far ahead.

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List of Abbreviations

BSF	Bunter Sandstone Formation
CO_2	carbon dioxide
LLF	Lincolnshire limestone formation
FPR	field average pressure
LGR	local grid refinement
CLDBOUNDARIES	all closed boundaries
BHP	bottom hole pressure
L.G	lithostatic gradient
LS	leman sandstone
TDS,	total dissolved solids
NTG	net to gross
GGR	global grid refinement
H2O	water
CL	chloride ion
WBHP	well bottom hole pressure
NACL	sodium chloride
XMF	Mole Fraction
CL^-	Chloride ion

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1. **AIM OF STUDY**.

The main aim of this research is to investigate the effect of pressurisation on the Bunter formation and how salinity evolves as brine migrates due to displacement by the CO_2 plume. The study also evaluates the impact of formation geometry on the brine migration behaviour and the variation of the salinity up dip.

1.1 Research Objective

The primary objective is to evaluate the potential risk of contamination of portable waters in the overlying Lincolnshire limestone aquifer caused by upward migration of brine through a legacy/leaky well and possible breakthrough of brine to the surface where the Bunter outcrops.

Results from this study would help in developing optimistic strategies for effectively injecting and storing CO_2 in deep saline formations, without causing significant environmental risks.

2. INTRODUCTION

2.1 Background

The geological storage of CO_2 in deep saline formations has become a popular strategy to eliminate the high amounts of greenhouse gases in the atmosphere. (Sally M. Benson et al, 2008), (K. Michael et al, 2010). The Bunter formation, deposited in the early Triassic era by fluvial and aeolian deposits (g), is regarded as a significant asset for CO_2 projects due to its excellent reservoir properties and high storage capacity. (Jon G. Gluyas and Usman Bagudu, 2020), (Hollinworth, A.D: et al., 2024).

The UK government awarded Shell and ExxonMobil a license, CS028, to inject CO_2 into the Bunter Sandstone Formation (BSF). Two saline aquifers are considered in this licensed area, which include the BSF and LS; however, this study focuses on the BSF.

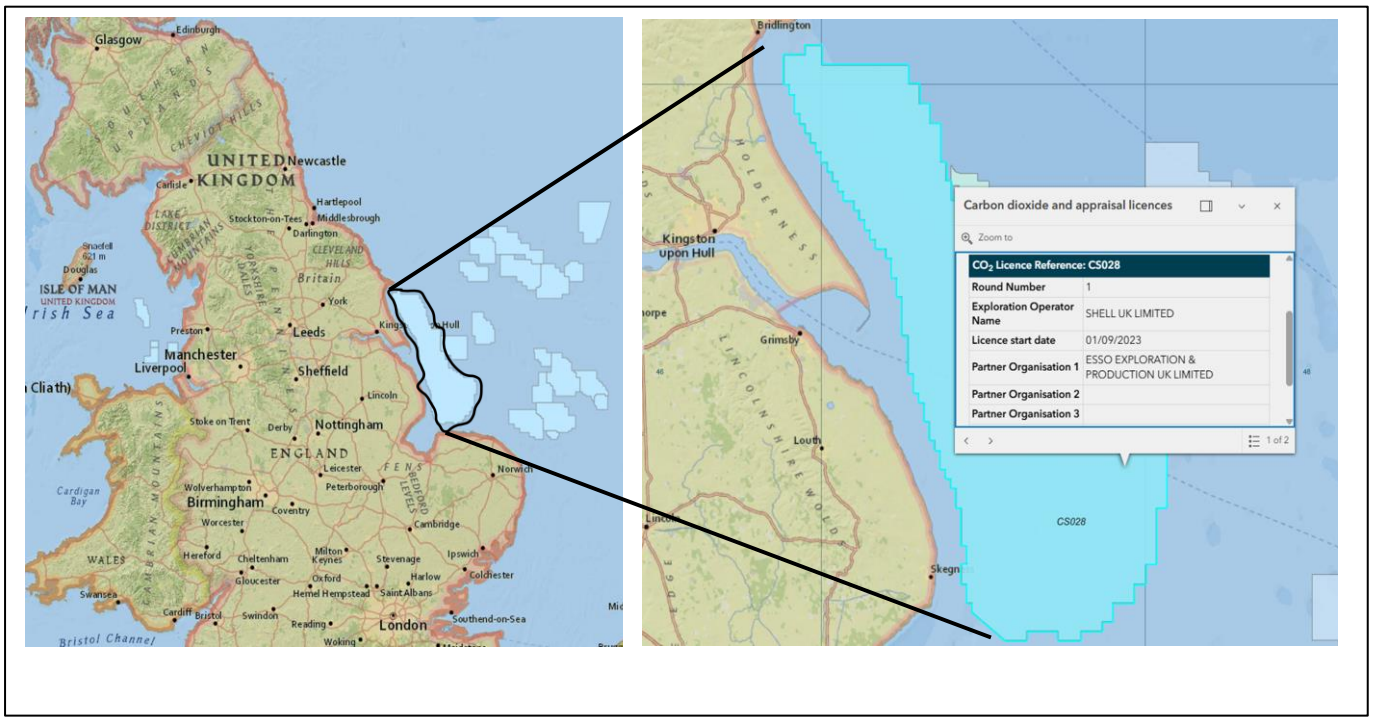

Figure 1: Map showing the licence 28 area (CS028), the area of focus for this study. ((NSTA), 2025)

The successful storage of CO_2 in deep saline formations is essential without causing any environmental risk to Shallow freshwater aquifers, as leaky wellbores are considered as potential pathways for CO_2 and brine leaks (Litang Hu,et al, 2012). The Bunter is unconfined

at its outcrop and is confined further offshore by the overlying Mercia Mudstone, which is highly impermeable (Mallin Martin, D., and P. L. Smedley, 2020). Therefore, the deeply confined layers are not frequently recharged with fresh water, thus exhibiting a higher salinity. If this very salty water migrates upwards due to pressurisation of the system, it could result in an increase in salinity as it encounters the portable waters located near where the bunter outcrops to the surface. (W.M Edmunds, and P.L Smedley, 2000).

2.2 Literature Review.

Carbon capture and storage in subsurface formations has become an optimized strategy to mitigate climate change due to the large amounts of emissions from industrial plants and oil and gas production (S. M. Benson et al, 2008). Deep saline aquifers are the most common formations in which CO_2 is injected because they have the highest storage capacity compared to other target formations. (Bachu & Shaw, 2003). The Bunter formation, located offshore in the Southern North Sea, has been considered a highly promising saline aquifer for CO_2 storage, with an estimated capacity of 5700 MCO_2t (Hollinworth, A.D: et al., 2024). It has excellent reservoir properties and is overlain by a sealing caprock. The Bunter Sandstone Formation (BSF) is equivalent to the Sherwood Sandstone Group located onshore along the western coast of the UK. (S.Hossain et al, 2024) . (S.Hossain et al, 2024) discovered 13 facies, 12 of which were observed from core samples and eight from the outcrop, suggesting lateral continuity of sediment deposition. Porosity values range from 15% to 24% and permeability varies widely, ranging from 0.1 to 5300md. For this study, the permeability of the Bunter will be taken as 500 md, as it is the median value of the observed permeability range across different facies. The Mercia Mudstone overlies the Bunter formation and serves as a regional seal, given its very low permeability. From the study of (P. J. Armitage. et al, 2015), it was discovered that Mercia has a permeability of 10^{-20} m^2, approximately 10^{-5}md, preventing vertical flow of CO_2 and

displaced brine in a pressurized system. CO_2 is injected into deep saline aquifers with salinity greater than 10,000 mg/L TDS, as this is not considered freshwater for use. (Assaf Wunsch et al., 2011).

CO_2 injection results in pressurization of the system, potentially causing displacement of brine within pore spaces, as studied by (Abdullah Cihan, et al, 2012). A study by (Jens T. Birkholzer, et al, 2009) revealed that pressure increases rapidly from the injector well, more than the CO_2 plume migrates. Therefore, if there are any leakage points, brine could flow significantly through the conduit path and contaminate shallow aquifers. Brine migration could cause serious environmental impacts if it reaches the surface through the open Bunter formation or migrates via leaky surfaces, such as a faulty well or a non-sealing fault, crossing into shallow groundwater aquifers (Abdullah Cihan, et al, 2012).

The Lincolnshire limestone is a widely active groundwater aquifer for domestic and commercial use. It was deposited between the lower Cretaceous and the upper Jurassic periods. It is fine-grained, with a thickness of 40m and outcrops at the west boundary. (J. P. Bloomfield et al, 2015). The aquifer is unconfined near the outcrop with the impermeable Mercia mudstone overlying it and becomes confined as it dips eastwards by both the Oxford and Mercia mudstones. (K J Griffiths,et al, 2006). High transmissibilities have been recorded for this formation as flow is mainly through the fractures, with the transmissibilities ranging from 260 to 2260m^2/day and with a geometric mean of 660m^2/day per (J. P. Bloomfield et al, 2015). The porosity and permeability of the LLF selected for the sensitivities in this study are 15% and 15md, respectively, based on the study by (R. B.Greswell et al, 1998) on the interpretation of the hydro-petrophysical properties of the LLF. If a leaky well that passes through the Lincolnshire Limestone Formation (LLF) into the Bunter Sandstone Formation (BSF), and its

integrity is compromised, the brine could flow vertically through weak cement bonds and contaminate groundwater in Lincolnshire.

Deep saline aquifers tend to have salinity gradients mainly due to residence time and the type of recharge system they possess. The salinity of the brine increases the longer it resides inside the formation. Brine at deeper depth has a longer interaction with the rock minerals, thus having a higher salinity than brine at shallower depth. A study by (W.M Edmunds, and P.L Smedley, 2000) reveals that through residence indicators, the residence time that the brine has interacted with the rock can be determined, thus determining the age and salinity difference of the brine with depth. The outcrop of the Sherwood sandstone is a recharge zone (A. H. Bath et al, 1987), and the flushing that occurs when groundwater is recharged from the western outcrop flows downgradient and mixes with the old brine, thus altering the chemistry. Dilution of the brine reduces the salinity, especially near the surface, where the Bunter sandstone is unconfined. Towards the eastern side, where it becomes confined by the Mercia and the underlying Bunter shale formation, the flushing effect on salinity diminishes as there is restriction of flow due to reduced permeability and the low permeability Mercia mudstone.

One of the highlights from (P.L. Smedley et al , 2018) shows that the Bunter in the East Midlands has the deepest fresh groundwater aquifer, with the oldest recorded at 500m. It is of Pleistocene age and has a chloride ion (Cl^-) concentration of 15 mg/l and a total dissolved solids of 500 mg/l. The salinity increases downgradient due to the interaction of the brine with gypsum, with chloride (Cl) concentrations rising to 5,000 mg/L, recorded at a depth of 800 m. The brine salinity at 2000m is taken as 303,000mg/l TDS, (E. A. Warren and P. C. Smalley, 1994)

3. **METHODOLOGY**.

3.1 Static Model Design

A 3D compositional model of the Bunter Sandstone Formation (BSF) was constructed using ECLIPSE 300 software. The model illustrates a gently dipping structure observed in the literature review (A. H. Bath et al, 1987)and extends 142 km from the western coast eastward, covering 52.7 km from north to south. Each of the cells in the grid block is 100m in both the x and y directions and is divided into eight vertical layers. Layers 2 and 5 are left active in the model as the sensitivities are carried out on the Lincolnshire formation and Bunter formation, as they are the main formations from the lithostratigraphic column being studied in this analysis. The other layers are made inactive to reduce the simulation time. The grid consists of 5,986,720 cells, with dimensions of 1420 × 527 × 8 cells in x, y, and z directions. A sub-model of the entire grid block is created and runs from cell 261 to 265 in the y direction, as shown in Figure 2. This model is termed the global grid in all the simulation models run in this study and is 0.5 km wide. This narrow strip is designed to analyze the worst-case scenario of pressure buildup in the system as the overlying Mercia mudstone is completely sealing off the BSF. A local grid refinement of the global grid strip is designed to allow for a finer resolution near the injector well and assumed leakage pathways, enabling detailed visualization of brine flow behavior and salinity changes across the two formations. The LGR strip is 300 m wide, with both the Lincoln and Bunter strips refined to 3 cells in the y direction; however, the Lincoln strip comprises five layers, while the Bunter strip has 50 layers in the z direction.

3.2 Modelling of Petrophysical Properties.

The porosity and permeability values of the Bunter formation are derived from core samples and sedimentological logs obtained from wells 42/25-1 and 42/25d-3. (S.Hossain et al, 2024). These core data values from this study characterize the permeability and porosity of the Bunter formation. Figure 21 in the appendix illustrates a vertical succession created based on facies

interpretations from the core samples. This lithofacies column is used to model the properties of the layers in the BSF strip within the design of the grid block in the methodology. Facies such as the planar cross-bedded sandstone (Sp), structureless sandstone (Sm), and planar cross-bedded sandstone with mud clasts (Spmc) exhibited the highest permeability, ranging from 9.5 to 5350 md. Notably, a high-permeability streak of 5300 md, from layer 11 to 15 of the BSF local grid, was modelled to extend laterally across the entire strip in the x direction. This was done to visualize the impact of a high-permeability zone on brine migration and salinity gradients.

To model the permeability anisotropy of the BSF, a kv/kh ratio of 0.1 was applied. This value is consistent with the range reported in the case study of (J.D.O. Williams et al, 2013), where kv/kh sensitivities between 0.01 and 0.36 were measured, for the fluvial–aeolian sandstones. (Clair Gough and Simon Shackley, 2005) study suggests that saline aquifers are completely saturated with brine in their pore spaces; therefore, the NTG of the field is taken as 1.

PARAMETER	**VALUE**	**REFERENCE**
GLOBAL GRID REFINEMENT SETTINGS		
porosity of BSF	0.2	(S.Hossain et al, 2024)
Porosity of LLF	0.15	(R. B.Greswell et al, 1998)
Permability of the LLF	0.15	(R. B.Greswell et al, 1998)
permeability of the BSF	500.0 md	(S.Hossain et al, 2024)
kv/kh	0.1	(J.D.O. Williams et al, 2013)
NTG	1k	(Clair Gough and Simon Shackley, 2005)
LOCAL GRID REFINEMENT SETTINGS		
high perm stream (BSF)	5300md; layer (11- 15): z direction)	(S.Hossain et al, 2024)

Table 1: summary of petrophysical properties of the BBF and LLF

3.3 Salinity Gradient Model.

Case studies by Edmund et al. (1982) and Smedley and Edmunds (2002) reveal that the brine in the Bunter exhibits a salinity gradient, increasing with depth. The salinity gradient designed for the Bunter narrow strip is represented in the table below. NaCl is modelled as the third component (Xmf_3) in the simulation, and its composition is traced over time to evaluate the change in salinity of the brine as it migrates away from the injector well.

Depth(m)	H2O	CO_2	NaCl	Measured CL- mg/L	Source/reference
	Xmf_1	Xmf_2	Xmf_3		
0	0.99997884	0	2.1158E-05	20	(W.M Edmunds, and P.L Smedley, 2000)
100	0.99998668	0	1.3321E-05	13	(W.M Edmunds, and P.L Smedley, 2000)
500	0.99998433	0	1.5672E-05	15	(W.M. Edmunds , et al, 1982)
671	0.99664896	0	0.00335104	3300	(A. H. Bath et al, 1987)
800	0.99492554	0	0.00507446	5000	((W.M Edmunds, and P.L Smedley, 2000)
1300	0.95578296	0	0.04421704	44000	(A. H. Bath et al, 1987)
2000	0.821613	0	0.178387	TDS=303,000	(E. A. Warren and P. C. Smalley, 1994)

Table 2: salinity gradient of the Bunter formation modelled for all cases

3.4 Model case design and sensitivities.

The simulation model cases were designed to investigate how the boundary conditions, shifting from an open system to a more confined system, affect pressure build-up within the BSF and, with this, evaluate the impact of this formation's geometry on brine migration and salinity evolution. Wells that are drilled in the Welton oil field, which was discovered in 1981 (James Steventon, 2018) are considered as legacy wells due to their completion in 1984. Welton Well is drilled through both the LLF and BSF formations; therefore, if it has weakened cement bonds, it could possibly create a leakage pathway for brine into the LLF. The Mercia mudstone is modelled to have a permeability of $1x10^{-5}$md as (P. J. Armitage. et al, 2015) reveals that the Mercia has a low permeability of 10^{-20} m^2 (10nD).

For all the cases, multiple cells within the local grid refinement of the BSF and LLF, and 10km apart, were selected to monitor the pressure propagation as the CO_2 plume migrates and salinity changes within the formations.

3.5 Completion of the injector well and leaky wells

The CO_2 injector well is completed in cell (1300i, 2j) in all 50 layers of the BSF. CO_2 is injected for 50 years in all the simulation scenarios, and the well is shut in for 150 years to monitor the plume migration and resultant brine displacement. The injector well is rate controlled with 1Mtpa and 2Mtpa injection rates applied in the different cases to analyse the effect of pressurisation within the BSF AND LLF. The leakage pathway for brine is simulated through two leaky wells. The Welton well, considered a leaky well, is located in cell (1100i * 2j) and is completed in all five layers of the LLF strip and 50 layers of the BSF strip. Another leaky well is modelled 40km away from the well and is completed only in the Bunter.

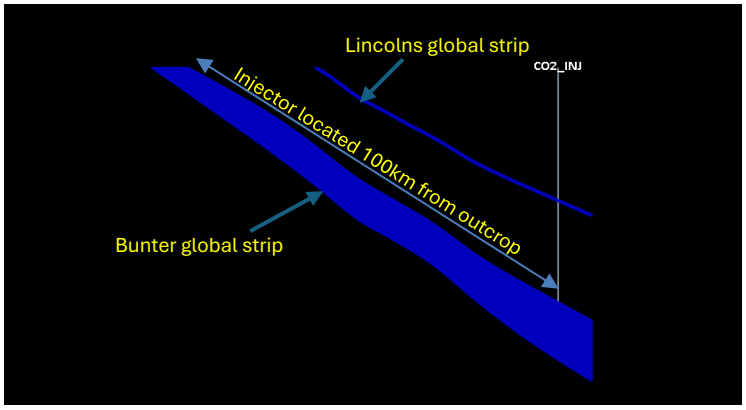

Figure 2: 2D schematic of the global grid of the Bunter and Lincoln formations; injector placed 100 km from the outcrop

4. SIMULATION RESULTS

In this section, the entities of each case are described, and the results are noted as observed. The results from the simulation in each case demonstrate the impact of the boundary conditions on the pressure distribution within the BSF and LLF. Salinity changes are observed across all models as the brine migrates up-dip. In some cases, there is leakage of brine into the overlying LLF, which results in a massive change in the initial salinity of the aquifer.

4.1 Case One (Base Case)

This case comprises an open system, where all the boundaries of the global grid are open in all directions with the multiplier set to 1e06. The bunter outcrop is taken to be open as described by (A. H. Bath et al, 1987). The injection rate is set at 1Mtpa for 50 years, with an observation period of 150 years to observe the plume migration within the BSF. Two leaky wells are included in this sensitivity to assess the impact of two leaky wells on brine migration.

The field average pressure (FPR) within the BSF remains almost constant with a 1-bar increase during the 50-year injection period and post shut-in period. The pressure in the global grid cells is highest 20km away from the injector, reaching over 212 bars by the end of the observation period; however, in other cells, there is a very low increase in pressure. The NaCl concentration of NaCl is tracked across selected cells to monitor changes in the TDS of brine, thereby interpreting the salinity variation across the BSF. Figure 5 shows the change in Xmf_3 during the injection and post-injection phases. Local grid cells selected to analyse the change in the Xmf3 as brine is displaced reveal that salinity near the injector in the upper layers gradually increases during the first two years of injection and becomes relatively constant in the following years post-injection. No leakage is observed in the Lincolnshire Limestone Formation (LLF) through the leaky well, as there is no pressure buildup noted in the aquifer.

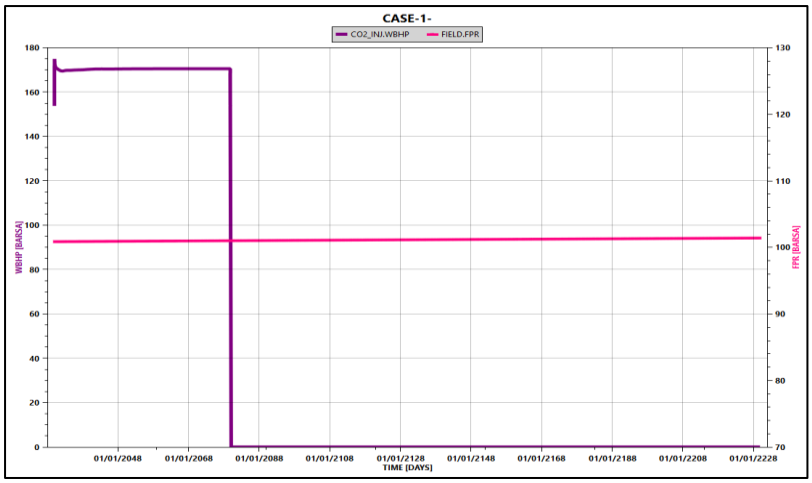

Figure 3: Field average pressure (FPR) and WBHP of the CO_2 injector over the injection and shut in period for case one.

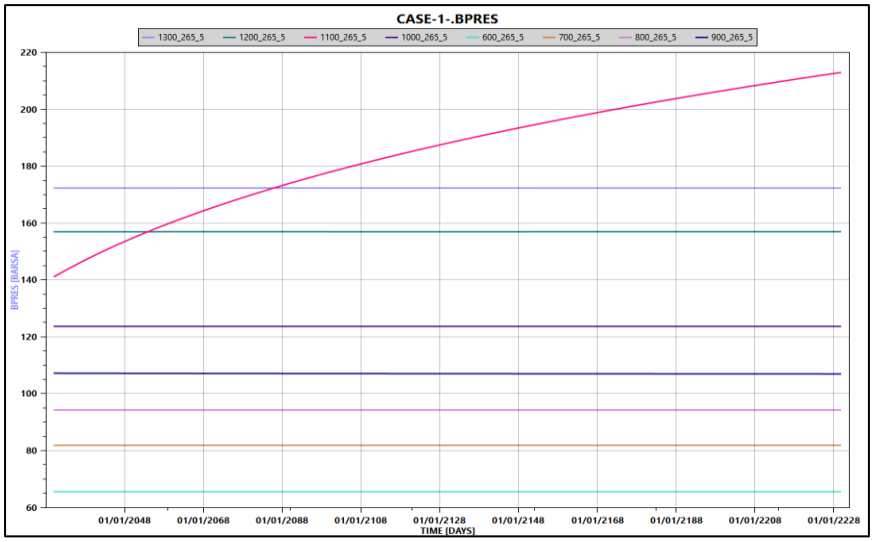

Figure 4: Pressure distribution across the monitored cells in the local grid refinement of the BSF in case 1-

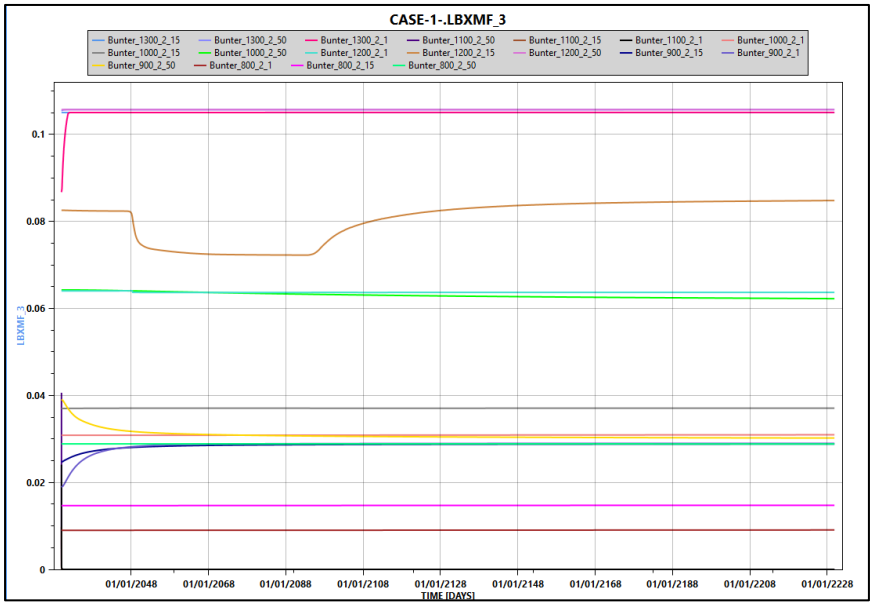

Figure 5: Change in the Xmf_3 across the LGR cells of the bunter

4.2 Sensitivities on the boundary transmissibility multipliers**.**

Variable transmissibility multipliers were applied to the global strip boundaries in all directions to evaluate the effect of boundary transmissibility on the pressurization of the BSF and the salinity of brine. The sensitivity multipliers selected include (1e04, 1e03, 1e02, 1e01), with an

injection rate of 50 Mtpa in all cases. The lower the multipliers, the more confined the BSF and the higher the pressure buildup within the system.

The highest increase in the field average pressure is observed in Case 1e01, reaching a peak of 136 bars, and later declines to 89 bars by the end of the post-injection phase, which lasts 34 years. The other three cases exhibit a gradual increase in the field pressure that later becomes constant during the shut-in period of 150 years. Brine leakage into the Lincolnshire Limestone Formation (LLF)was observed through the leaky wells, which is not observed in the base case, where the boundaries are fully open. The large crossflow of brine into LLF is observed in the cases with lower multipliers, with the highest amount of leakage occurring in case 1e01. Figure 6 illustrates the extension of the brine leakage within the LLF for different multipliers. Salinity within the LLF changes drastically following the brine leakage. The highest salinity change is observed in case 1e01, where the Xmf_3 concentration in the local grid cells of the LLF increases rapidly compared to the other scenarios.

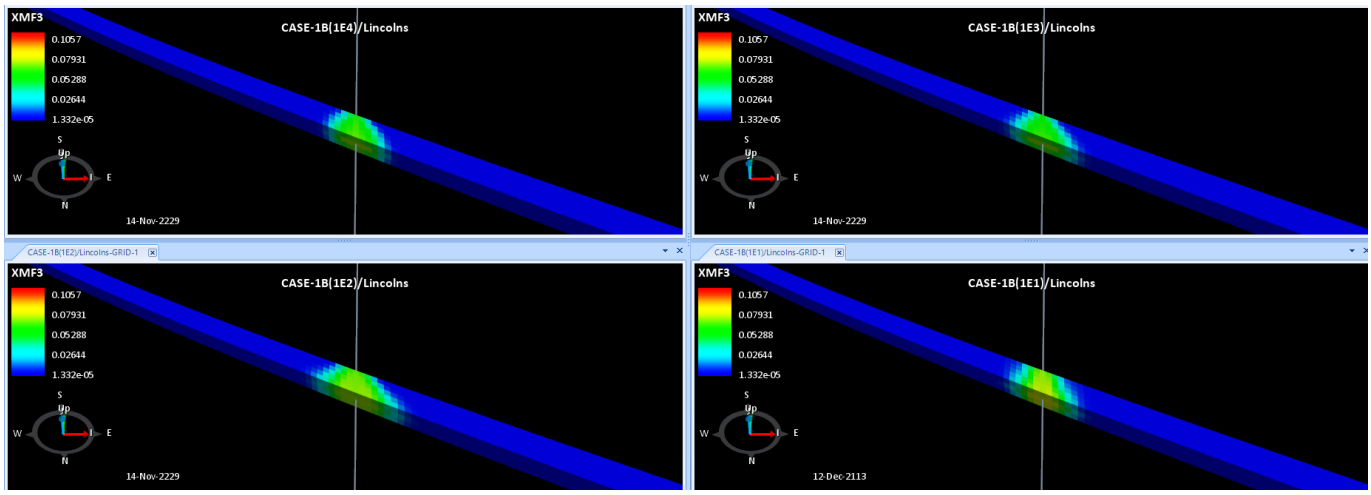

Figure 6: schematic of the variation in the brine leakage into the Lincolns formation with variation in the transmissibility multipliers applied on the boundaries

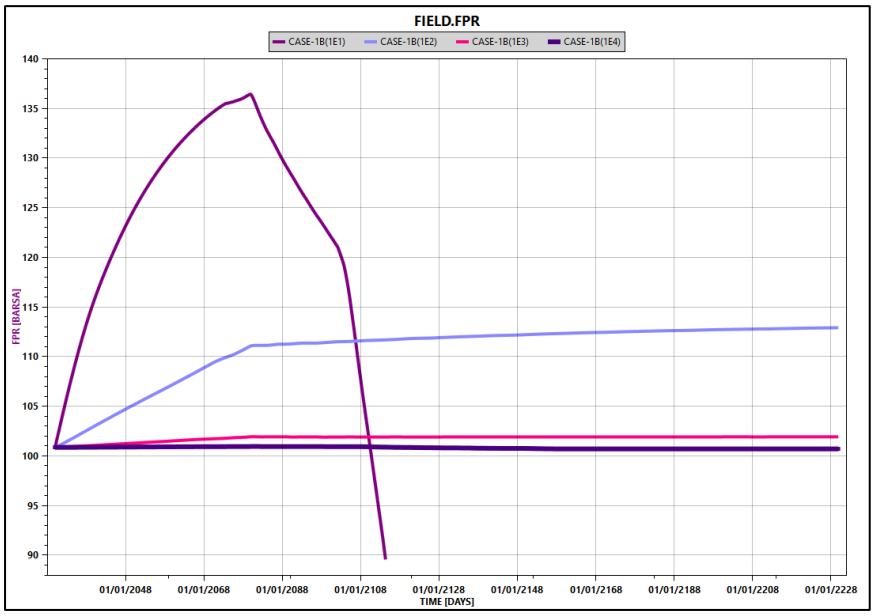

Figure 7: Field average pressure (FPR) VS time in cases 1e01, 1e02, 1e03, 1e04; increased pressurization of the bunter compared to the open boundaries scenario in case one.

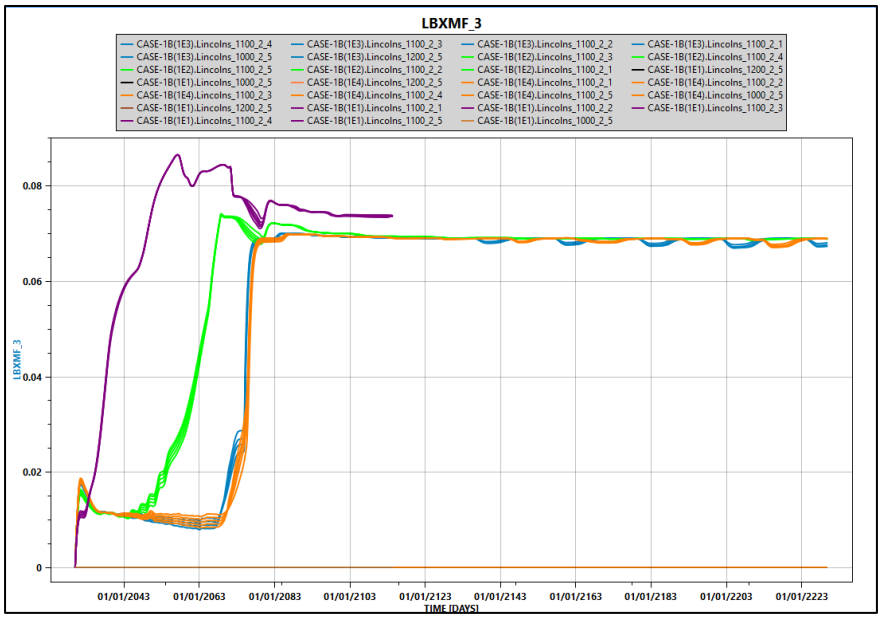

Figure 8: salinity variation in the Lincolnshire formation following the brine leakage into the freshwater aquifer for cases (1e01, 1e02, 1e03, 1e04)

4.3 Case Two (Worst Case Leak into Lincolnshire Limestone Formation).

This model evaluates the effect of a closed system, where all the boundaries in all directions of the global grid are closed; however, the Lincolnshire Limestone Formation (LLF) outcrop is assumed to be open with a 1e06 multiplier applied to the western boundary. The injection rate in this case is increased to 2 Mtpa for 50 years, followed by a shut-in period of 99 years. This

sensitivity is considered the worst-case scenario of a leak into the LLF because a high-pressure buildup is expected as more CO_2 is injected into the BSF. This could result in significant displacement of brine up dip and upward migration of the brine through the leaky well.

From the monitored cells, the localized pressure across the BSF increased rapidly, with the highest pressure observed nearest to the injector, reaching 410 bars by the end of the injection period. The pressure gradually declines during the shut-in period. A gradual increase in the pressure of the LLF is observed within the first year of injection, indicating that brine upward migration through the leaky well occurs immediately the injection phase proceeds. Figure 9 illustrates the pressure distribution within the LLF during the 50 years of injection till the end of the observation period. While salinity within the BSF remains relatively constant, the LLF shows a rapid increase in salinity once the salty water mixes with the groundwater.

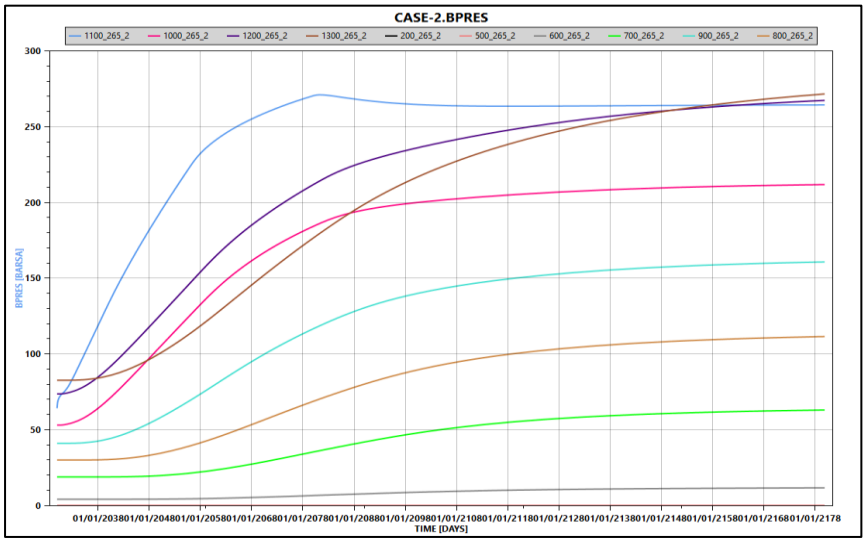

Figure 9: pressurisation of the LLF in case 2

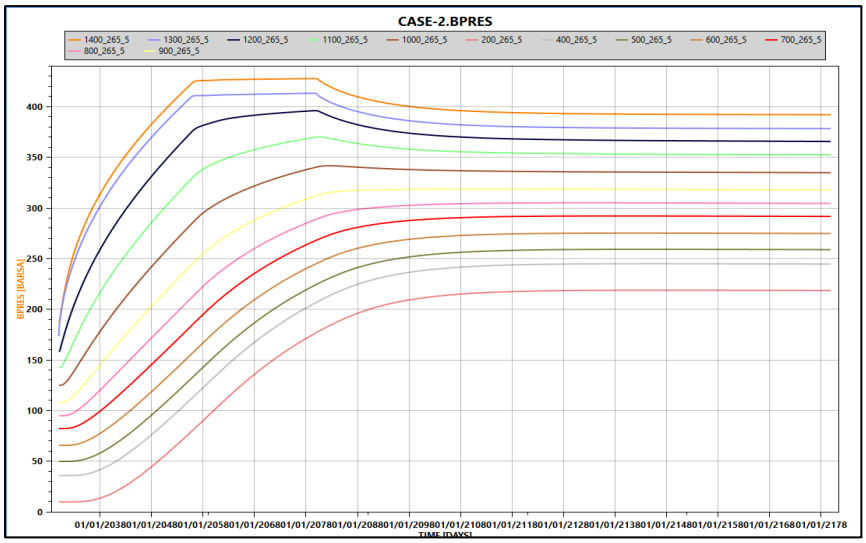

Figure 10: pressure build up monitored from the cells within the bunter local grid.

4.4 Case 2B (All boundaries are closed, including the LLF outcrop).

Case 2b has similar sensitivity parameters as in case two; however, the key difference is that the LLF outcrop in this case is assumed to be completely sealed. This model is designed to assess the effect of a fully closed system on the upward migration of brine and salinity change within the LLF. Although it is not realistic that the boundaries in the actual BSF and LLF field are completely sealed, this scenario provides a worst-case estimate of the amount of leakage and change in salinity with a highly pressured system. 50 Mt of CO_2 is injected over 50 years, followed by a shut-in period of 150 years. The extended observation period in this case is designed to evaluate the long-term impact of a fully closed system on the pressurisation of both the LLF and BSF, and to determine the extent of brine leakage.

From the simulation results, there is a higher pressure buildup within the BSF compared to case two, which is modelled with an open LLF outcrop. This causes a larger volume of brine to flow upwards via the leaky well into the LLF. This results in a higher change in salinity as more salty water is pushed into the aquifer. Figure 11 shows the comparison of the concentration of Xmf_3 around the point of leakage for cases 2 and 2B (all boundaries closed).

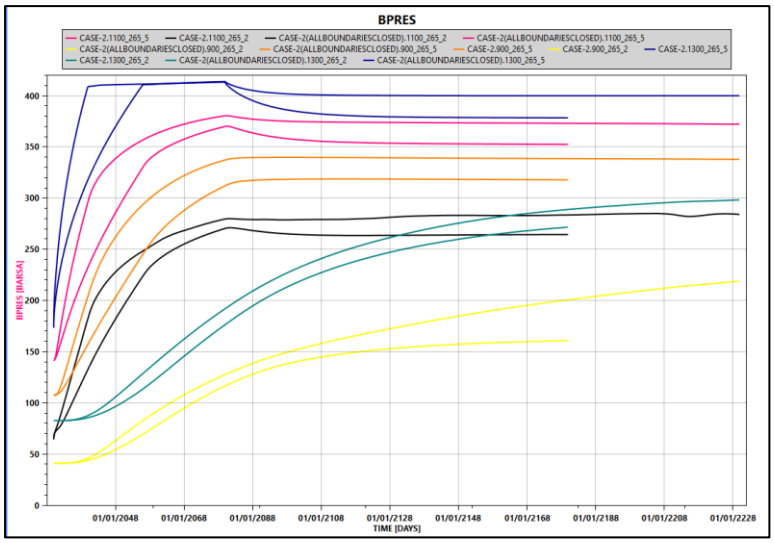

Figure 11: Comparison of pressurisation within the BSF for Case 2 and Case 2b(all boundaries closed). As multiple cells within the local grid are visualised, the comparison cells from each case are plotted in a single colour for a simplistic interpretation.

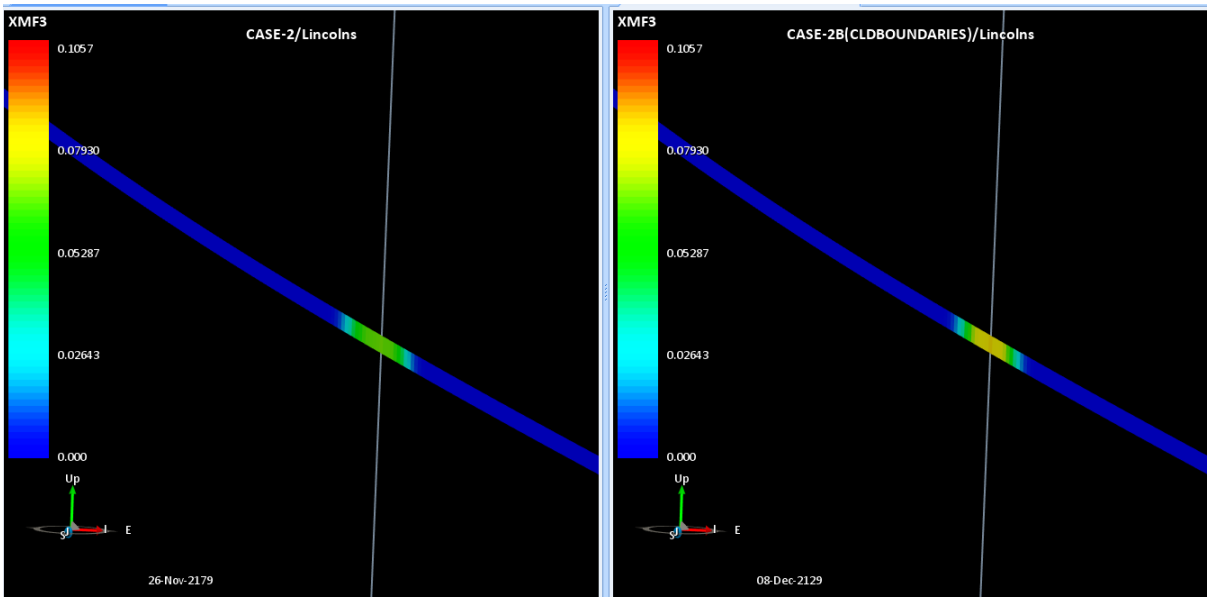

Figure 12: 3D schematic diagram illustrating the comparison of the amount of salty water intrusion into the liconls formation between case two and case 2 (CLDBOUNDARIES)

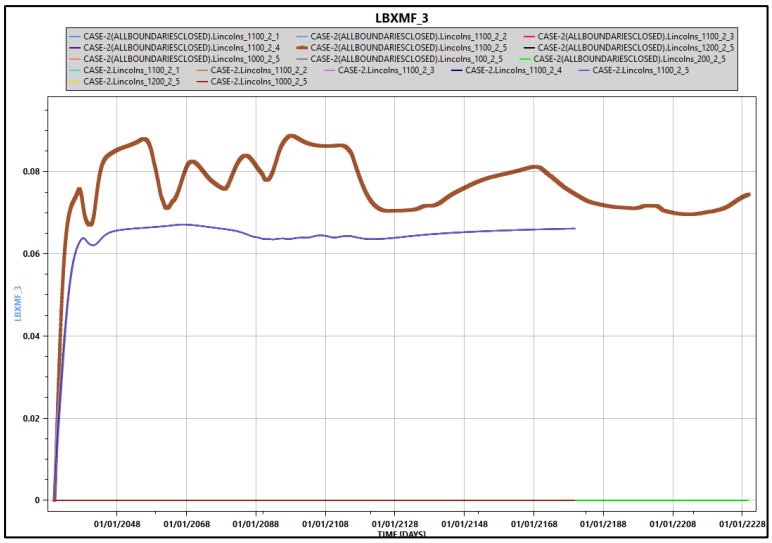

Figure 13: 2D plot of the Xmf$_3$ in local grid cells of Lincolns with time, the change in salinity is demonstrated as the highest variation *occurring in case2(CLDBOUNDARIES)*

4.5 Case Three (No Leaky Wells).

In this case, the global grid boundaries are closed, but the BSF outcrop is modelled as open (A. H. Bath et al, 1987). No leaky wells are included in the model. The purpose of this sensitivity is to analyse the extent of brine migration within the BSF up-dip and to assess the potential breakthrough to surface through the open outcrop. 1 Mtpa CO_2 is injected for 50 years, and the well is shut-in for 150 years.

Following the injection of 50 MtCO_2 over 50 years, pressure within the BSF increases rapidly, particularly between cells 1300 and 900, which corresponds to 40 km from the injector. The highest pressure observed is 422 bar in the cell (1300i * 2j * 5z) in which the injector well is completed. When the well is shut in after 50 years of injection, the pressure gradually declines as the propagation continues across the whole formation. By the end of the shut-in period, residual brine saturation is only observed in the upper layers, extending 64 km from the injector well, within the top 14 layers of the Bunter, and the brine saturation in the lower layers does not change significantly. The saturation plot in Figure 16 illustrates how brine saturation changes during the injection and post-injection phases, showing the amount of brine displaced

as the CO_2 plume primarily migrates in the upper layers despite being injected into all 50 layers of the BSF. The salinity variation is observed in the upper layers, where significant displacement of brine occurs. There is a gradual increase in salinity during both the injection and shut-in periods, which stabilizes to almost constant levels, with insignificant changes in Xmf_3 noted.

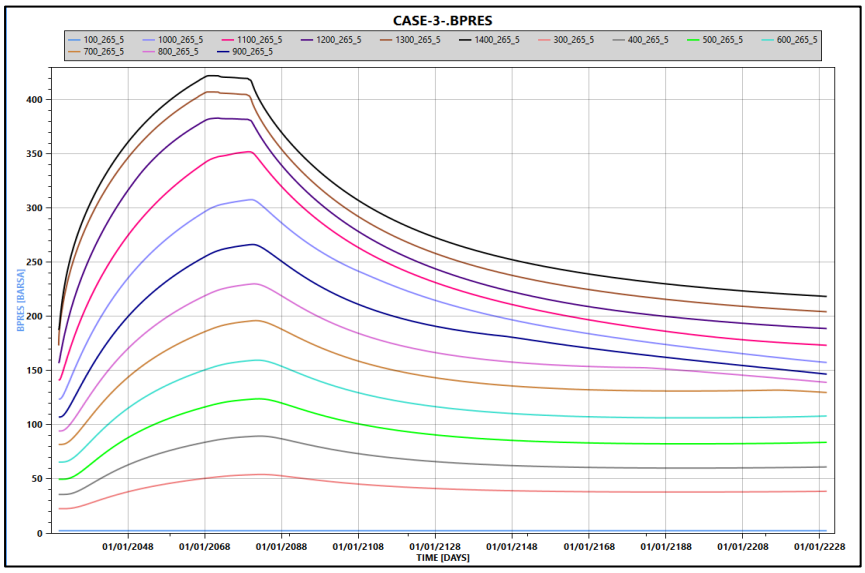

Figure 14: pressure build up within the LGR of the bunter

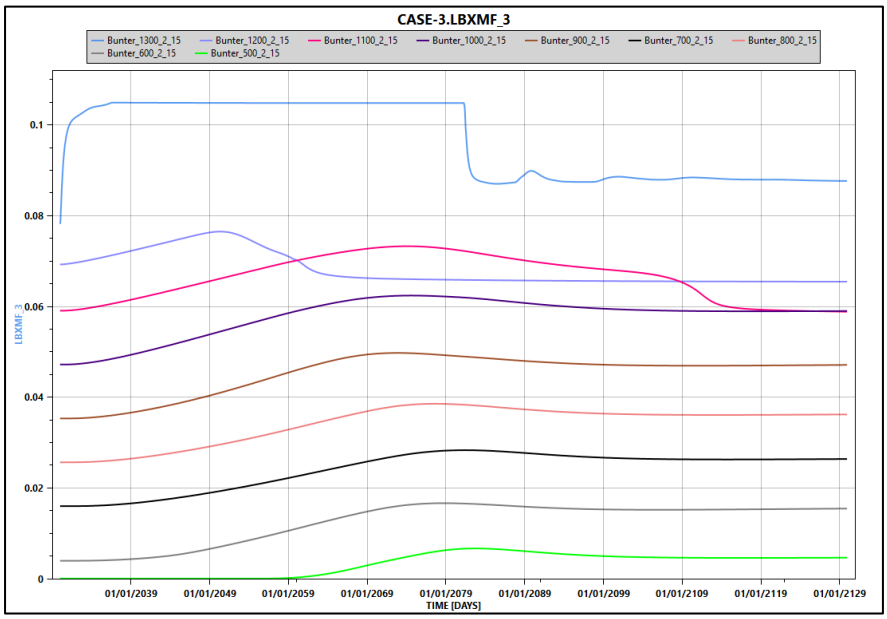

Figure 15: change of Xmf_3 with time within the bunter for case 3.

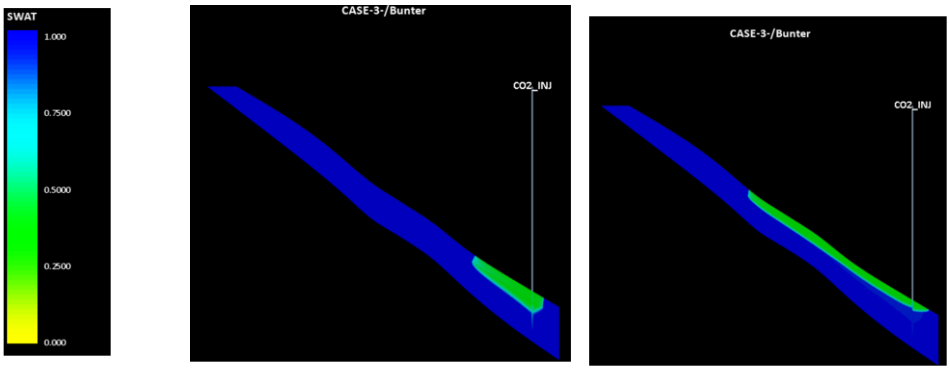

Figure 16: change in brine saturation following the injection of 50Mtpa of CO_2 in 50 years and shut-in period of 150 years in case 3-.

5. DISCUSSION OF RESULTS.

5.1 Effect of BSF pressurisation on Brine migration and leakage.

The high pressurisation of the BSF is associated with closed boundaries because the pressure buildup following the CO_2 injection of 100Mtpa for case 2 and 2b and 50Mtpa for case 3 is constrained and does not get to spread out.

This causes the BSF to become overpressurized. The study by (Assaf Wunsch et al, 2013) described that invasion of brine from deep saline formations into groundwater aquifers is possible if the hydrostatic head in the saline aquifer is higher than that in the shallow groundwater aquifer, provided there is a conductive pathway created by a leaky well or a non-sealing caprock. The high-pressure buildup in the BSF, caused by closed boundaries, results in a high-pressure drawdown where the brine travels from the highly pressurized BSF to the low pressured LLF through the leaky well. The higher the pressure in the BSF, the larger the intrusion of salty water into LLF, as seen in case 1e01 and case 2B(allboundariesclosed). In case one (base case), where the transmissibility multiplier is set to 1e06 at the boundaries, the reservoir acts as an open system, allowing pressure to disperse quickly across the BSF during and after the injection period. This results in a reduced drawdown, decreasing the driving force for the brine to migrate up dip. This explains why no leakage of brine into the LLF is observed, as the brine cannot flow upwards through the leaky well due to the low-pressure gradient

between the BSF and LLF. Significant displacement of brine is observed in the upper layers, with residual brine mainly visualized in the top 17 layers of the BSF local grid, and the farthest displacement front at 60 km from the injector well. Once the CO_2 is injected into the BSF, the plume rises upward because CO_2 is less dense than the brine, displacing the brine in those upper regions. Significant displacement of brine is noted in the cases where the multipliers were varied. Large brine movement is observed in the cases where the boundary multipliers are varied, with the magnitude of brine migration increasing as the lower boundary multipliers are applied. The high permeability streak influences the preferential movement of the CO_2 plume in the upper layers, causing the large displacement of brine in those zones. There is still uncertainty about the lateral connectivity of these facies; however, assuming it is continuous toward the western outcrop provides a conductive pathway for brine movement, which could lead to breakthrough through the BSF outcrop that is presumed to be open (A. H. Bath et al, 1987)

5.2 Salinity variation

Generally, there is a slight increase in brine salinity observed across all sensitivities. Salinity change is mainly observed in the upper layers of the BSF, with a gradual increase within the first year of injection that later stabilizes over the subsequent years. This results from the minimal brine displacement observed within the BSF, as the saltier water from the deeper layers does not migrate far and mix with the less saline water in the respective layers up dip. The brine in the BSF comprises a salinity gradient (W.M Edmunds, and P.L Smedley, 2000) thus, once the salty water from the deeper layers mixes with the fresh water in the LLF aquifer, it quickly increases its salinity. The change in salinity across the LLF is controlled by the level of the brine displacement. The NACL concentration (Xmf_3) of NaCl within the selected cells was recorded, and the equivalent brine salinity in ppm was calculated to illustrate the change in salinity.

In the base case, in the high permeability zone in cell (1200i * 2j* 15z), 20 km from the injector, the salinity appears to remain constant but rapidly decreases during the seven years from the start of the injection phase. This could likely be caused by CO_2 dissolution, which increases the overall volume of brine in that region. The same number of dissolved solids would then be distributed in a larger volume of brine, causing a reduction in salinity. Salinity later increases 15 years post shut in and stabilizes by the end of the shut-in period. From the other cases, 30 to 50 km from the injector, high salinity changes are observed mainly in the high perm layers of the BSF as the saltier brine preferentially moves in these regions due to high reservoir connectivity. There is no variation in salinity recorded at 60km and beyond from the injector, as saltier brine does not migrate this far within the model.

5.3 Limitation of the BHP on Injection Rate.

There were scenarios where the bottom hole pressure of the injector well was choked back, reducing the injection rate. This is observed in the cases where the boundaries are closed and the BSF is over pressured. This indicates that in situations with closed boundaries, it is not possible to sustain injecting 2Mtpa in case 2 and 2 (all boundaries) or 1Mtpa in case 3 for a period of 50 years. This is limited by the BHP of 400 bars designed for the injector well. The injection rate in the simulation is reduced because of the limiting BHP. Less CO_2 is injected into the BSF, which results in less brine being displaced. Therefore, no major changes in salinity will occur. The BHP limit of 400 bar is used in all cases of this study, mainly as a sensitivity parameter. The lithostatic gradient of the Bunter is given as 337 bars, based on the case study of (ref). In the ideal BSF field, the optimal BHP limit for the injector wells would be set at 303 bar, which is 90% of the lithostatic pressure. Considering that a higher injection pressure has been applied to the Bunter formation and salinity changes are minimal due to negligible brine movement, operating conditions with a limit of 303 bars would likely result in even less brine migration and minor salinity changes than those observed in this study.

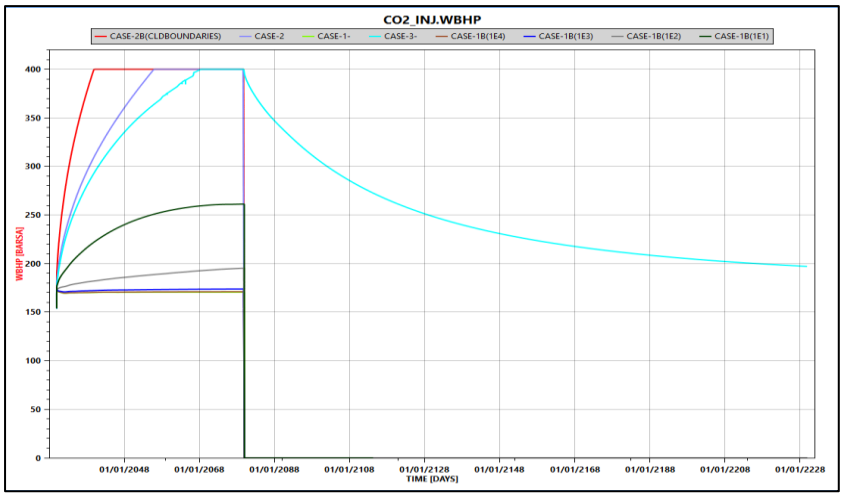

Figure 17: showing the cases whose injection rate is cut back due to the limitation of the BHP; the flat line at 400 bars shows the limit with which the pressure at the injector rises.

5.4 CO_2 plume migration and leakage.

A leak of CO_2 is observed in the LLF, with the sensitivities involving a variation in the boundary multipliers. In a less-pressurized system, the CO_2 plume exhibits a lower density because the lower pressurisation allows for the expansion of the gas, thus occupying a larger volume within the BSF. This results in a larger migration of the CO_2 plume as observed in the (1e01, 1e02, 1e03, 1e04). Figure 18 illustrates the CO_2 plume migration patterns for cases where brine leakage was observed. No CO_2 leakage within the Lincolnshire Limestone Formation (LLF) is observed in cases 2 and 2B, as the plume does not reach the Welton well, which is located 20 km from the injector. This suggests that a high-pressure buildup within the system, caused by the confinement of the boundaries, leads to a high CO_2 density, as it becomes compressed and occupies a smaller volume within the BSF. Consequently, this results in less migration of the plume up dip. In case 1e04, an increased flow rate of brine from the second leaky well is noted, following the breakthrough of CO_2 in the second leaky well. This suggests that an artificial lift effect stimulated by the CO_2 gas increases the brine flow rate through the conductive pathway of the leaky wells. The CO_2 plume migration limits the movement of the salty water plume within the LLF. Therefore, the amount of CO_2 leakage influences the extent of the brine displacement front from the leaky well.

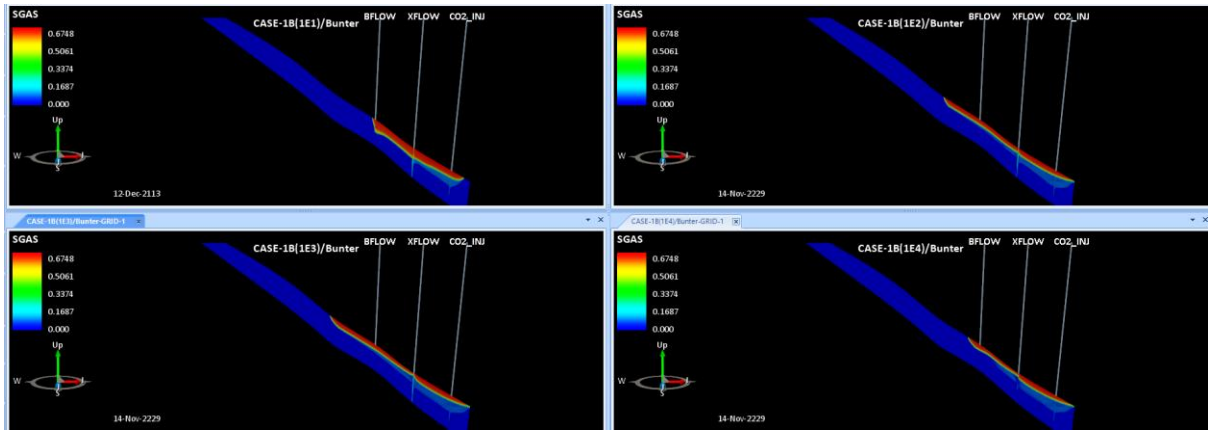

Figure 18: 3D schematic of the saturation of the gas illustrating the CO_2 plume migration behavior within the Bunter formation for the cases with varying transmissibility multipliers.

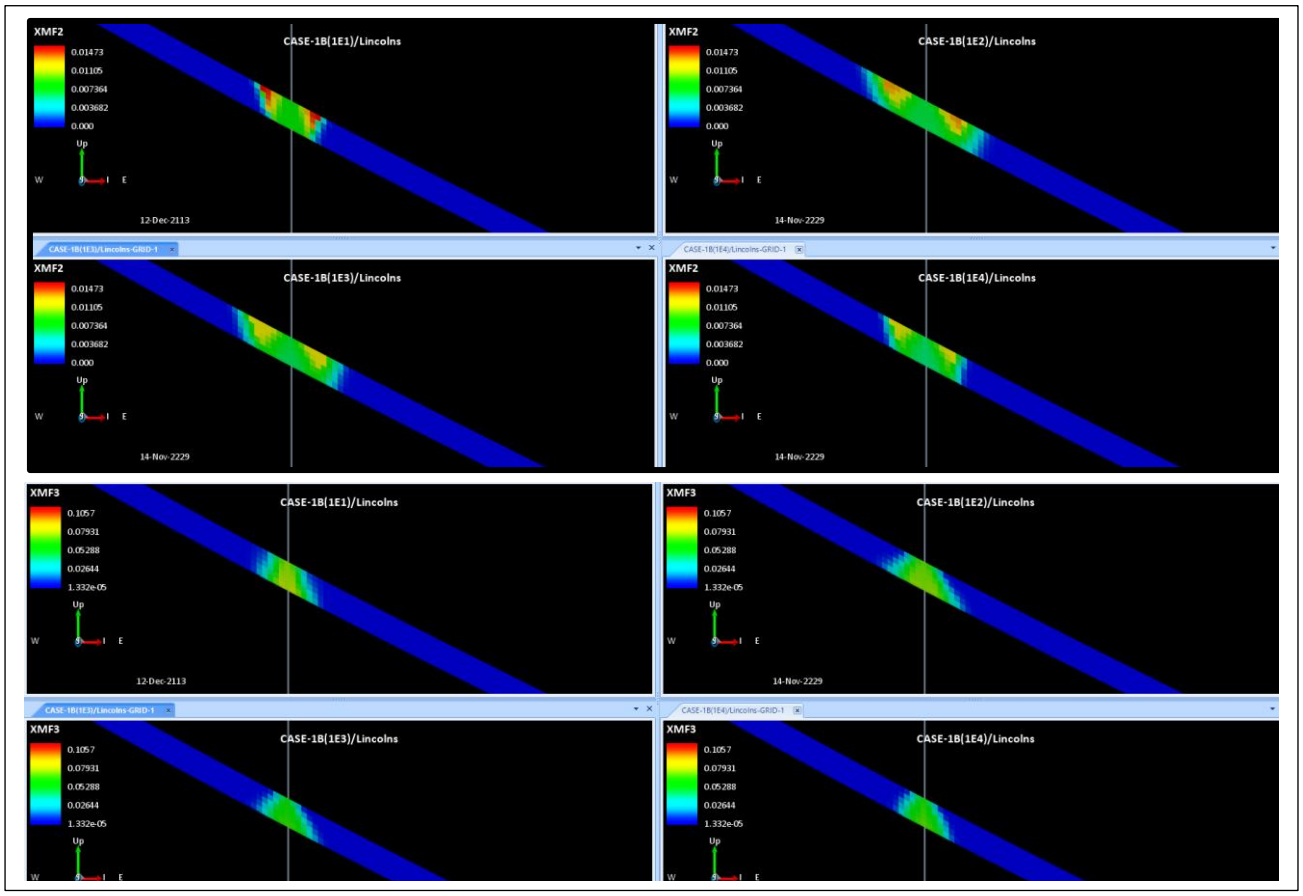

Figure 19: Illustration of the influence of the CO_2 plume in the Lincolns formation on the brine migration from the leaky well

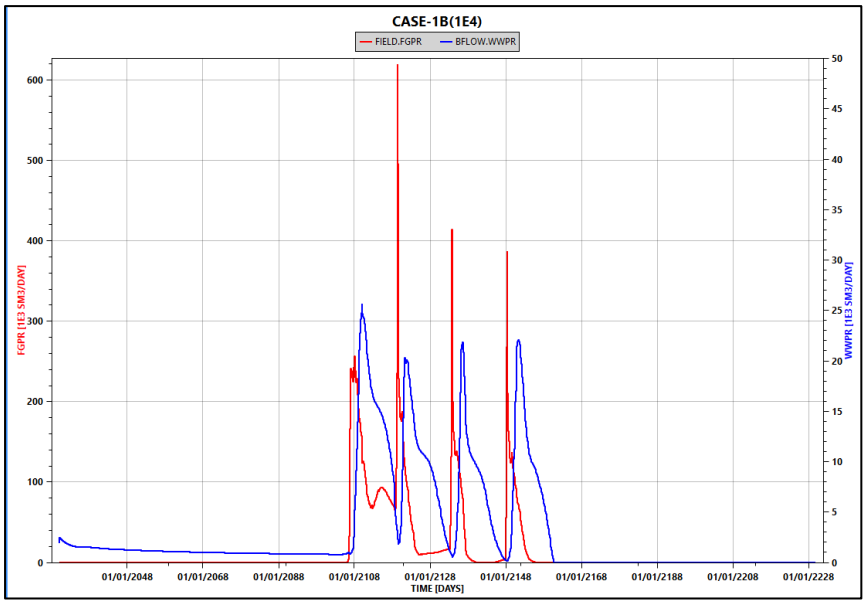

Figure 20: The plot illustrates how the rate of brine leakage increases as the breakthrough of CO2 increases.

6. CONCLUSIONS.

This study examined sensitivities on boundary conditions to evaluate their effect on the pressurization of the BSF located in the license CS028 area, offshore eastern England coast. Following the pressurisation of the saline aquifer, the brine migration and salinity changes were analysed to assess the potential risk of brine crossflow into the shallow groundwater aquifer located in the overlying LLF and leakage through the BSF outcrop.

A static reservoir simulation model was used to design multiple cases that include variable boundary conditions and leakage scenarios via leaky wells. Although there is uncertainty regarding the lateral continuity of the planar cross-bedded sandstone (Sp), facies within the BSF, a high-permeability streak of 5300md (S.Hossain et al, 2024) is assumed to be continuous throughout the LGR of the Bunter to investigate the impact on the salinity change.

Open boundary conditions observed in case one (base case) resulted in low pressurisation of the BSF. This results in minimal brine displacement and a slight increase in salinity. When the transmissibility of the boundaries decreases, there is increased pressurization within the BSF, which causes more extensive displacement of brine. The brine residual front in these scenarios

is observed at 62 km from the injector, which is 43 km more than the residual brine front seen in case one.

The worst-case scenario of brine leakage into the LLF occurs when the bunter is most confined, as the transmissibility of the boundaries is lowered, or when they are entirely closed as observed in case 2B.

The high pressure build-up within the BSF creates a high-pressure drawdown, allowing for the upward flow of brine through the leaky well. This occurs because the BSF is highly pressurized compared to the LLF. (Assaf Wunsch et al, 2013)

The salinity of the brine in the BSF is negligible due to minimal brine migration. Significant changes are observed in the high-permeability layers, where brine preferentially migrates because of higher reservoir connectivity. A considerable increase in salinity occurs in the LLF once the saltier brine from the BSF contaminates the fresh water in the LLF. This poses a high environmental risk since the Lincolnshire limestone aquifer is a major source of water for domestic and commercial use.

In conclusion, the chance of large amounts of salty water being pushed upward into the LLF through the leaky well and migrating more than 1.5 km is low. The system is exaggerated by modeling completely closed boundaries, which is very unlikely since the Bunter formation is discovered to be open (A. H. Bath et al, 1987). Additionally, there is still uncertainty about the status of Western Lincoln's outcrop, whether it is open or closed. Therefore, with the optimal injection rates limited by a BHP of 303 bars, the system would become pressurized as CO_2 is injected; however, less brine would migrate uphill towards the outcrop or flow upwards via potential leakage wells into overlying portable aquifers.

7. RECOMMENDATIONS.

Based on the simulation results, the following recommendations are suggested for the safe and effective injection and storage of CO_2 in the Bunter Sandstone Formation (BSF). The presence of a legacy well within the target formation poses a significant risk to the overlying aquifer. Regular maintenance of these wells should be performed to ensure that brine does not flow vertically into the overlying shallow aquifers. No significant brine migration was observed during a 200-year simulation; therefore, it is highly unlikely that brine would break through the open Bunter outcrop. However, pressure monitoring during CO_2 injection operations should be conducted regularly, as over-pressurization of the formation could lead to substantial brine migration. Observation wells could be drilled to monitor brine salinity during plume migration. Quality characterization of the target formation should be carried out to identify high-permeability zones or leaking faults that could lead to elevated brine migration.

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9. APPENDIX

	low perm range(md)	high perm range(md)
FI	0.18	2.92
SSS	1.14	588
SH	7.2	2468
ST1,	35	1673
ST	13	2813
SP	9.5	5350
SM	9.5	5350
SPMC	4	3853
GMG	0.7	89
SI	35	1673

Figure 21: vertical succession of the lithofacies discovered in the Bunter by Hosain et al

DESCRIPTION	**CASE-1** Base case with everything included	**CASE-2** Worst leak in Lincolnshire	**CASE-3** No leaky wells
Assume that relative to this strip:			
North boundary	OPEN	CLOSED	CLOSED
South boundary	OPEN	CLOSED	CLOSED
East boundary(fault)	OPEN (leaking fault)	CLOSED	CLOSED
CO_2 injection rate of	1 Mtpa	2 Mtpa	1 Mtpa
CO_2 Injection well @ cell	1300	1300	1300
Lincolnshire Limestone Formation	YES	YES	NO
Lincolnshire outcrop	OPEN	OPEN	CLOSED
Bunter Sandstone Formation (BSF)	YES	YES	YES
Bunter outcrop	OPEN	CLOSED	OPEN
4.5D high perm streak in Bunter	YES	YES	YES
Variable salinity gradient in Bunter	YES	YES	YES
Constant salinity in Lincolnshire	YES	YES	
Leaky well from Bunter to surface @ cell	900	NO	NO
Leaky well from Bunter to Lincolnshire Limestone @ cell	1100	1100	NO

Table 3: summary of the sensitivities carried out in this research study